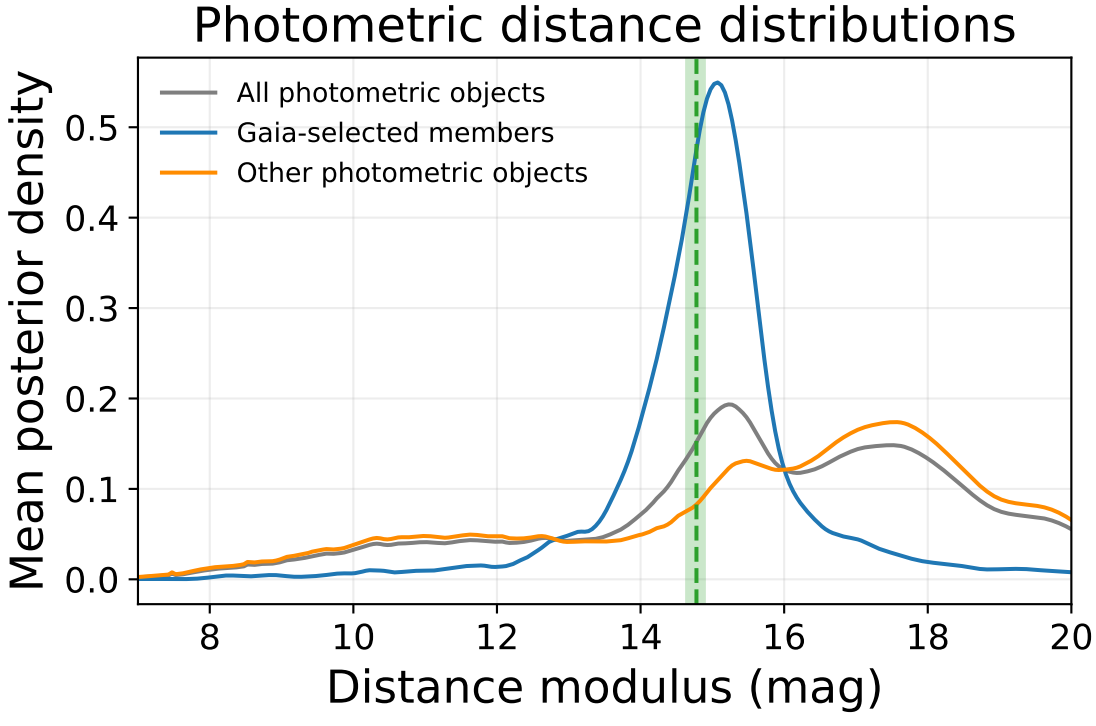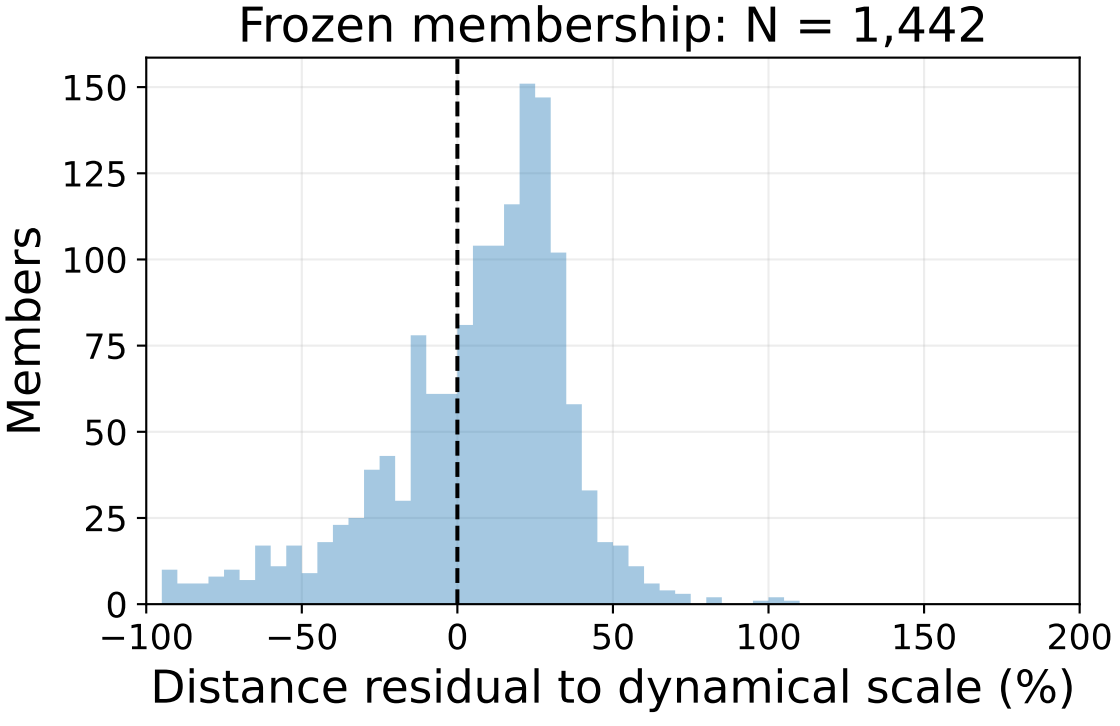NGC 288: marginalization and an independent distance scale